

Joseph Rance

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Education

University of Oxford

DPhil in Engineering Science

2025-

- Researching test-time scaling in language models
- Supervisors: Philip Torr, Adel Bibi

University of Cambridge

MEng in Computer Science

2021-2024

- Distinction in all modules and dissertation
- Dissertation: *Cost-Aware Drug Design by Optimising Backward Policies of GFlowNets*
- Supervisors: Miruna Cretu, Pietro Liò

BA in Computer Science

2021-2024

- First Class in all three years, prize for highly commended dissertation
- Dissertation: *Evaluating attacks on fairness in Federated Learning* (link)
- Supervisors: Filip Svoboda, Nicholas Lane.

Publications

Towards Better Exploration in Sequential Test-Time Scaling (link)

Under review

Joseph Rance, Fabio Pizzati, Juil Sock, Woody Bayliss, Marc Górriz Blanch, Philip Torr, Adel Bibi

Can Private Machine Learning Be Fair? (link)

The 39th Annual AAAI Conference on Artificial Intelligence, 2025

Joseph Rance, Filip Svoboda

Augmentation Backdoors (link)

ICLR 2023 Workshop on Backdoor Attacks and Defenses in Machine Learning

Joseph Rance, Yiren Zhao, Ilia Shumailov, Robert D. Mullins

Cost-Aware Drug Design by Optimising Backward Policies of GFlowNets (link)

Dissertation, 2025

Joseph Rance

Evaluating attacks on fairness in federated learning (link)

Dissertation, 2024

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Experience

Software Engineering Intern, Microsoft

Summer 2023

- Designed and implemented upgrades to the Azure-for-Operators MLOps codebase for a 75% cost reduction by processing low-priority data during off-peak times.
- Advocated for a more general implementation that can be made open-source (link).
- Led a student team during the Microsoft global hackathon to train a reinforcement learning agent to optimise server scheduling on Azure.

Research Intern, University of Cambridge

Summer 2022

- Developed three new data-poisoning attacks hidden in data augmentation functions.
- Presented our paper (see above) at the ICLR BANDS workshop.
- Supervisors: Yiren Zhao, Ilia Shumailov, Robert Mullins.

Student Volunteer, AI4Good organisation

Summer 2020

- Worked as part of a team to simulate the spread of coronavirus in refugee camps.
- Produced metrics to evaluate the simulation, which was used to inform real-world decision making.

Awards & Achievements

Awards:

- ESPRC IDLA Research Studentship
- Robinson College Scholarship
- CST Department Award for Highly Commended Part II Dissertation
- Arkwright Engineering Scholarship
- 4x Cambridge full blue, 1x Oxford full blue

Last updated: 04/10/2026